

An Algorithm for Integrated Structure and Control Design with Variance Bounds

Guojun Shi¹ and Robert E. Skelton

*SSCL, 1293 Potter Engineering Center,
Purdue University, West Lafayette, IN 47907-1293, shig@ecn.purdue.edu*

Abstract

This paper presents an approach to integrate structure and control design. The objective of the problem is to find: i) the optimal values of the structure parameters, *e.g.*, stiffness, damping ratios, and actuator location parameters, *etc.*; ii) an optimal stabilizing state feedback controller such that the active control energy is minimized subject to: a) the prespecified RMS constraints on the outputs, and b) the constraints on structure parameters. The algorithm is given in terms of LMI's, and the convergence is guaranteed. The solution is optimal in the "local sense". An example is given to illustrate the effectiveness of the algorithm.

1 Introduction

System design technology has taken on a more interdisciplinary nature. This has been caused by more demanding performance criteria and design specifications of all types of machines and structures in various fields, *e.g.*, semiconductor manufacturing, civil engineering, automobile engineering, *etc.* Passive control alone may not meet the high specifications. On the other hand, pure active control may be very expensive to realize. This has led researchers to integrate the passive and active control design in a certain optimal sense to satisfy the high demanding performance requirements. See [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12].

The modeling and control problems are not independent[13]. The structure design and control design are not separable and necessarily are iterative. This paper proposes an iterative algorithm to integrate structure and control design.

The phrase "passive control" is used to mean a redesign the structure parameters, *e.g.*, mass, stiffness, damping ratio, to produce a more desirable response. It may be necessary in many instances to alter the structure by adding vibration absorbers or isolators in order to meet vibration criteria.

Active control, on the other hand, is characterized by a closed-loop system associated with actuators, sensors and the structure. It relies on real time measurements of the response of the system. It differs from the passive control in its ability to inject energy into the system, as well as store and dissipate it.

The structure parameters are adjustable within specified bounds. This fact puts boundaries on dynamic performance of the system achievable by passive control. So under certain stringent design specifications when passive control can not ensure the desired performance, the introduction of active control along with the passive control constitutes a viable alternative. The combination of the two methodologies in a certain optimal sense can achieve better performance than simply designing the controller after the structure is designed, and better performance than either structure redesign (passive control) or active control acting alone. One can benefit greatly by coordinating both structure and active control redesign. Unfortunately, these theories have not been integrated in any tractable way.

An algorithm is proposed herein to integrate the structure and control design. Specifically, the algorithm simultaneously finds: i) the optimal values of the stiffness, damping ratios, and actuator location parameter, and ii) an optimal stabilizing state feedback controller such that the active control energy is minimized subject to: a) the prespecified RMS constraints on the outputs, and b) the constraints on the structure parameters. This problem is referred to as the Optimal Mix of Passive and Active Control (OMPAC). The algorithm provides a systematic approach to tune the structure parameters and design an active controller.

The paper is organized as follows. Section 2 gives the problem formulation. Based on the vector-second order structure model, the OMPAC problem is stated. For the initial structure, the active controller design is accomplished in Section 3. Section 4 investigates the structure-controller redesign problem. In Section 5, the (X, K, p) - Design Loop is considered. In this loop, the optimal controller and structure parameters in stiffness and damping matrices are obtained with fixed actuator location parameter δ . In Section 6, the problem

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of selecting the actuator location parameter δ is considered. The algorithm to integrate the structure and control design is present in Section 7. The convergence of the algorithm is guaranteed. Section 8 gives a numerical example to illustrate the effectiveness of the algorithm. Finally, some conclusions are given.

The notation used in this paper is fairly standard. For a matrix A , A^T denotes its transpose. For symmetric matrices P and Q , $P > 0$ ($P \geq 0$) denotes the fact that P is positive definite (positive semidefinite), $P > Q$ ($P \geq Q$) denotes $P - Q > 0$ ($P - Q \geq 0$). $\|A\|$ denotes the largest singular value of A . $\mathcal{E}_\infty[\cdot] \triangleq \lim_{t \rightarrow \infty} [\cdot]$, and $\mathcal{E}$ is the expectation operation.

2 Problem Formulation

Consider a preliminary linear time-invariant dynamical model for a structure with the following vector second-order representation

$$\mathcal{M}\ddot{q} + \mathcal{D}(\alpha)\dot{q} + \mathcal{K}(\beta)q = \mathcal{B}_1 w + \mathcal{B}_2(\delta)u \quad (1)$$

$$z = \mathcal{C}_1 q + \mathcal{C}_2 \dot{q} \quad (2)$$

$$y = \begin{bmatrix} q \\ \dot{q} \end{bmatrix} \quad (3)$$

where $q \in \mathbb{R}^{n_s}$ is the displacement vector, $\dot{q}$ and $\ddot{q}$ are the velocity and acceleration vectors, respectively, $u \in \mathbb{R}^{n_u}$ is a control input vector, $z \in \mathbb{R}^{n_z}$ is the output vector to be regulated, $y \in \mathbb{R}^{n_y}$ is a vector of measurements, $w \in \mathbb{R}^{n_w}$ is the white noise with intensity $W > 0$, uncorrelated to the initial system state $[q^T(0) \ \dot{q}^T(0)]^T$, and

$$\mathcal{D}(\alpha) = \mathcal{D} + \Delta\mathcal{D}(\alpha) = \mathcal{D} + \sum_{i=1}^d \alpha_i D_i \quad (4)$$

$$\mathcal{K}(\beta) = \mathcal{K} + \Delta\mathcal{K}(\beta) = \mathcal{K} + \sum_{j=1}^k \beta_j K_j \quad (5)$$

$$\mathcal{B}_2(\delta) = \mathcal{B}_2 + \Delta\mathcal{B}_2(\delta) = \mathcal{B}_2 + \sum_{l=1}^b \delta_l B_l \quad (6)$$

where $\alpha = [\alpha_1 \ \alpha_2 \ \dots \ \alpha_d]$ is the set of system parameters that can be designed to adjust the system damping, $\beta = [\beta_1 \ \beta_2 \ \dots \ \beta_k]$ is the set of system parameters that can be designed to adjust the system stiffness, $\delta = [\delta_1 \ \delta_2 \ \dots \ \delta_b]$ is a vector of actuator and location parameters, the matrices $\{D_i\}_1^d$, $\{K_j\}_1^k$, and $\{B_l\}_1^b$ are the corresponding basis matrices. The matrices $\mathcal{M}$, $\mathcal{D}$, $\mathcal{K}$, $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{C}_1$ and $\mathcal{C}_2$ are constant matrices with appropriate dimensions that represent the nominal structure design. Also

$$\alpha \in [\underline{\alpha}, \bar{\alpha}], \beta \in [\underline{\beta}, \bar{\beta}], \delta \in [\underline{\delta}, \bar{\delta}]$$

where $\underline{\alpha}$, $\bar{\alpha}$, $\underline{\beta}$, $\bar{\beta}$, $\underline{\delta}$, and $\bar{\delta}$ are specified constants that represent the structure design constraints.

Define $p = [\alpha \ \beta]$. The state space representation for the dynamical system (1)-(3) is given by

$$\begin{aligned} \dot{x} &= A(p)x + B_1 w + B_2(\delta)u \\ z &= C_1 x \\ y &= x \end{aligned} \quad (7)$$

where

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \quad A(p) = A + \Delta A(p)$$

$$A = \begin{bmatrix} 0 & I_{n_q} \\ -\mathcal{M}^{-1}\mathcal{K} & -\mathcal{M}^{-1}\mathcal{D} \end{bmatrix}$$

$$\Delta A(p) = \begin{bmatrix} 0 & I_{n_q} \\ -\mathcal{M}^{-1}\Delta\mathcal{K} & -\mathcal{M}^{-1}\Delta\mathcal{D} \end{bmatrix}$$

$$B_2(\delta) = B_2 + \Delta B_2(\delta)$$

$$B_2 = \begin{bmatrix} 0 \\ \mathcal{M}^{-1}B_2 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 \\ \mathcal{M}^{-1}B_1 \end{bmatrix}$$

$$\Delta B_2(\delta) = \begin{bmatrix} 0 \\ \mathcal{M}^{-1}\Delta B_2(\delta) \end{bmatrix}$$

$$C_1 = [\mathcal{C}_1 \ \mathcal{C}_2]$$

$$p = [\alpha_1, \dots, \alpha_d, \beta_1, \dots, \beta_k]$$

$$\underline{p} = [\underline{\alpha}_1, \dots, \underline{\alpha}_d, \underline{\beta}_1, \dots, \underline{\beta}_k]$$

$$\bar{p} = [\bar{\alpha}_1, \dots, \bar{\alpha}_d, \bar{\beta}_1, \dots, \bar{\beta}_k]$$

Notice that the system matrix $A(p)$ is a linear function of p . The matrices A , B_1 , B_2 , C_1 , D_{12} are real constant with appropriate dimensions. We shall make the following assumptions for the system (1)-(3).

Assumptions

For any $p \in [\underline{p}, \bar{p}]$ and $\delta \in [\underline{\delta}, \bar{\delta}]$,

- (i) The pair $[A(p), B_2(\delta)]$ is stabilizable;
- (ii) The pair $[A(p), B_1]$ is stabilizable.

Remark 2.1 For discussions about the above assumptions, the reader is referred to [14].

In this paper, we consider the following problem.

The OMPAC Problem: Consider the structure (1). Given a performance vector

$$\{\sigma_1, \sigma_2, \dots, \sigma_{n_z}\},$$

find a stabilizing state feedback controller $u = Gx$, the structure parameters $p \in [\underline{p}, \bar{p}]$ and the actuator location parameters $\delta \in [\underline{\delta}, \bar{\delta}]$ such that the control effort $\mathcal{E}_\infty(u^T u)$ is minimized subject to the output variance constraint $\mathcal{E}_\infty(z_i^2) < \sigma_i$, $i = 1, 2, \dots, n_z$.

The OMPAC problem will be approximated by two subproblems: (i) choose the values for parameter vectors, p and δ to minimize the control effort subject to the output variance constraint, (which is passive design); (ii) design a controller G such that the control effort is minimized subject to the output variance constraint, (which is the active design). The two subproblems are not independent and should be done iteratively.

It will become clear that OMPAC problem is not convex. Hence, the global optimal point is difficult to find. In this paper, an alternating algorithm is proposed to find a local optimal point. In the next section, we shall consider the active control design problem.

3 Output Variance Constraint Control Design

In this section, the output variance constraint control problem is considered for the active control design. The system is the nominal plant, $[A, B_1, B_2, C_1, W]$.

Specifically, consider a system of (7) with fixed p and δ in A and B , respectively.

$$\begin{aligned}\dot{x} &= Ax + B_1w + B_2u \\ z &= C_1x\end{aligned}\quad (8)$$

Given a set performance scalars

$$\{\sigma_1, \sigma_2, \dots, \sigma_{n_z}\}$$

for system (8), find a stabilizing controller $u = Gx$ such that the resulting closed-loop system is asymptotically stable and the control effort $\mathcal{E}_\infty(u^T u)$ is minimized subject to

$$\mathcal{E}_\infty(z_i^2) < \sigma_i, \quad i = 1, 2, \dots, n_z.$$

The closed-loop system associated with system (8) and a stabilizing controller $u = Gx$ is then given by

$$\begin{aligned}\dot{x} &= (A + B_2G)x + B_1w \\ z &= C_1x\end{aligned}\quad (9)$$

Then we have the Lyapunov inequality for the closed-loop system

$$(A + B_2G)X + X(A + B_2G)^T + B_1WB_1^T < 0 \quad (10)$$

where $X > 0$ is an upperbound for the state covariance. Then the output variance constraint (OVC) control problem is to find a stabilizing $u = Gx$ such that $\text{trace}(GXG^T)$ is minimized subject to $[C_1XC_1^T]_{ii} < \sigma_i$, $i = 1, 2, \dots, n_z$. Then it is straightforward to show the following result.

Theorem 1 Consider the nominal system (8) and let the performance scalars

$$\{\sigma_1, \sigma_2, \dots, \sigma_{n_z}\}$$

be given. Suppose that the Assumptions in Section 2 are satisfied for the nominal system. Then the OVC state feedback control problem is equivalent to the following problem:

$$\min_{(X, K, \gamma)} \gamma, \quad \text{subject to} \quad (11)$$

$$\begin{bmatrix} \gamma I & K \\ K^T & X \end{bmatrix} > 0 \quad (12)$$

$$\begin{aligned}AX + XA^T + B_2K + \\ K^TB_2^T + B_1WB_1^T < 0\end{aligned}\quad (13)$$

$$[C_1XC_1^T]_{ii} < \sigma_i, \quad i = 1, 2, \dots, n_z \quad (14)$$

The optimal solution is denoted as (X_n, K_n, γ_n) , and the optimal controller is given by $G_n = K_nX_n^{-1}$.

The OVC control problem with the above formulation is a standard LMI problem and can be solved using LMITOOL [15].

4 (K, p) - Optimization

In this section, we consider the optimal selection of the structure parameter, p , and new controller design in the local sense. Suppose the solution of the OVC control problem, (X_n, K_n, γ_n) , is given for the nominal system (8). Now, let the state covariance upperbound X_n be fixed. Then we consider the following Lyapunov inequality

$$\begin{aligned}A(p)X_n + X_nA(p)^T + B_2K + \\ K^TB_2 + B_1WB_1^T < 0.\end{aligned}\quad (15)$$

with output variance $C_1X_nC_1^T$. Notice that the output variance constraints are automatically satisfied in this redesign phase, since from the OVC problem $[C_1X_nC_1^T]_{ii} < \sigma_i$, $i = 1, 2, \dots, n_z$.

Theorem 2 Suppose the state covariance upperbound $X_n > 0$ is given from Theorem 1. The (K, p) - Optimization problem is as follows:

$$\min_{(K, p, \gamma)} \gamma, \quad \text{subject to} \quad (16)$$

$$\begin{bmatrix} \gamma I & K \\ K^T & X_n \end{bmatrix} > 0 \quad (17)$$

$$\text{Inequality (15)} \quad (18)$$

$$p < \bar{p} < \bar{p} \quad (19)$$

The optimal solution is denoted as $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$. The optimal controller given by $G_{n+1} = K_{n+1}X_n^{-1}$ yields $\gamma_{n+1} \leq \gamma_n$.

Remark 4.1 As can be seen from Theorem 2 that the (K, p) - Optimization gives the local optimal value for the structure parameter, p , and a new controller, G . Furthermore, the control effort is nonincreasing.

5 (X, K, p) - Design Loop

Now, we consider the following subproblem. For fixed parameter, δ , $(B_2(\delta))$, find (X, K, p) such that the control effort $\mathcal{E}_\infty(u^T u)$ is minimized subject to the Lyapunov inequality and the output performance requirements, i.e.,

$$A(p)X + XA(p)^T + B_2K + K^T B_2^T + B_1WB_1^T < 0 \quad (20)$$

$$[C_1XC_1^T]_{ii} < \sigma_i, \quad i = 1, 2, \dots, n_z \quad (21)$$

The above subproblem can be solved iteratively as follows:

(X, K, p) - Design Subproblem: Consider system (7) with fixed δ . Let $\epsilon > 0$ be the given numerical tolerance.

- Step 0 - Set $p_0 = 0$, $n = 0$;
- Step 1 - Solve the OVC problem with fixed p_n by Theorem 1; Denote the solution as (X_n, K_n, γ_n) ;
- Step 2 - With X_n , solve the (K, p) - optimization problem to get $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$;
- Step 3 - If $|\gamma_n - \gamma_{n+1}| > \epsilon$, Go to Step 1; Otherwise, output $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$.

Notice that the solution of the (X, K, p) - Design Loop, is only a suboptimal solution in the sense that δ is fixed. In the next section, we consider the iteration between δ and the (X, K, p) - Design Loop.

6 (X, δ) - Optimization

In this section, we consider the selection for the actuator-type and location parameter δ . Let the solution of the (X, K, p) - Design Loop, $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$, be given for the system with fixed δ , say δ_n . Then the problem of (X, δ) - Optimization is to find new value for δ and new controller such that the control effort is minimized subject to the output variance constraints. The problem can be formulated as follows.

The (X, δ) Design Subproblem: Let the solution, $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$, to the (X, K, p) - Design Loop be given. Then the (X, δ) - Optimization problem is

$$\min_{(X, \delta, \gamma)} \gamma, \text{ subject to} \quad (22)$$

$$\begin{bmatrix} \gamma I & K_{n+1} \\ K_{n+1}^T & X \end{bmatrix} > 0 \quad (23)$$

$$A(p_{n+1})X + XA(p_{n+1})^T + B_2(\delta)K_{n+1} + K_{n+1}^T B_2(\delta)^T + B_1WB_1^T < 0 \quad (24)$$

$$[C_1XC_1^T]_{ii} < \sigma_i, \quad i = 1, 2, \dots, n_z \quad (25)$$

$$\underline{\delta} < \delta < \bar{\delta} \quad (26)$$

The optimal solution is denoted as $(X_{n+1}, \delta_{n+1}, \gamma_{n+2})$. Then $\gamma_{n+2} \leq \gamma_{n+1}$.

Remark 6.1 The (X, δ) - Optimization problem gives the optimal value for the parameter δ and a new controller, $K_{n+1}X_{n+1}^{-1}$, satisfying the performance requirements with less control effort.

7 The Algorithm to Integrate Structure and Control Design

In the previous sections, alternating optimization problems are considered. We are now in the position to give the an algorithm to integrate structure and control design. The basic idea is to iterate the (X, K, p) - Design Loop and the (X, δ) - Optimization Loop such that the control effort is monotonically nonincreasing.

Algorithm 1 Consider the system (7). Let the output performance requirements $(\sigma_1, \sigma_2, \dots, \sigma_{n_z})$ be given. Set $p_0 = 0$, $\delta_0 = 0$, $n = 0$, and the numerical tolerance $\epsilon > 0$.

- Step 0 - Solve the (X, K, p) - Design Loop with fixed δ_n to get $(X_n, K_{n+1}, p_{n+1}, \gamma_{n+1})$.
- Step 1 - Solve the (X, δ) - Optimization problem to get $(X_{n+1}, \delta_{n+1}, \gamma_{n+2})$.
- Step 2 - If $|\gamma_{n+1} - \gamma_{n+2}| > \epsilon$, goto Step 0. Otherwise, output $(X_{n+1}, K_{n+1}, p_{n+1}, \delta_{n+1}, \gamma_{n+2})$.
- Step 3 - The controller gain is $G = K_{n+1}X_{n+1}^{-1}$.

Theorem 3 In Algorithm 1, the control effort is monotonically nonincreasing and converges to a limit $\gamma_\infty \geq 0$.

Remark 7.1 If the limit $\gamma_\infty = 0$, then the problem reduces to passive control only. Therefore, Algorithm 1 also provides an approach to passive control design.

Remark 7.2 The results given in this paper can be easily extended to the more general case where $B_1(\phi)$ is a function of an adjustable free parameter ϕ .

8 A Numerical Example

Consider the lumped mass, fully connected system. The nominal values for the system parameters are:

$$m = \begin{bmatrix} m_1 & m_2 & m_3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 10 \end{bmatrix}$$

$$k = \begin{bmatrix} k_1 & k_2 & k_3 & k_4 & k_5 & k_6 \end{bmatrix} =$$

$$\begin{bmatrix} 2 & 2 & 4 & 1 & 3 & 1 \end{bmatrix} \\ d = \begin{bmatrix} d_1 & d_2 & d_3 & d_4 & d_5 & d_6 \end{bmatrix} = \\ \begin{bmatrix} 0.42 & 0.22 & 1.04 & 0.01 & 0.03 & 0.01 \end{bmatrix}.$$

A state space representation of the system is given by

$$\begin{aligned} \dot{x} &= A(p)x + B_1 w + B_2 u \\ z &= C_1 x \end{aligned}$$

where

$$\begin{aligned} A(p) &= \begin{bmatrix} 0 & I \\ -\mathcal{M}^{-1}\mathcal{K} & -\mathcal{M}^{-1}\mathcal{D} \end{bmatrix} \\ B_1 &= I, \quad B_2 = \begin{bmatrix} 0 \\ \mathcal{M}^{-1} \end{bmatrix} \\ C_1 &= I, \quad W = I \end{aligned}$$

and

$$\begin{aligned} \mathcal{M} &= \begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \\ \mathcal{K} &= \begin{bmatrix} k_1 + k_4 + k_5 & -k_4 & -k_5 \\ -k_4 & k_2 + k_4 + k_6 & -k_6 \\ -k_5 & -k_6 & k_3 + k_5 + k_6 \end{bmatrix} \\ \mathcal{D} &= \begin{bmatrix} d_1 + d_4 + d_5 & -d_4 & -d_5 \\ -d_4 & d_2 + d_4 + d_6 & -d_6 \\ -d_5 & -d_6 & d_3 + d_5 + d_6 \end{bmatrix} \end{aligned}$$

Suppose that we have the freedom to redesign the values for the spring and damper elements of the system. That is, the plant redesign parameters are

$$p = \Delta \begin{bmatrix} k_1 & k_2 & \dots & k_6 & d_1 & d_2 & \dots & d_6 \end{bmatrix}$$

where Δ means the variation around the nominal value. The open-loop output variances are $\{12.9187 \ 11.5764 \ 9.9833 \ 13.7206 \ 18.0466 \ 5.9361\}$. The performance requirements are

$$\begin{bmatrix} \sigma_1 & \sigma_2 & \dots & \sigma_6 \end{bmatrix} = [1 \ 2 \ 1 \ 2 \ 2 \ 2].$$

For this example we only need to solve the (X, K, p) - Subproblem, since B_2 is constant.

The constraints on the redesign parameters are given by

$$\begin{aligned} \underline{p} &= \begin{bmatrix} -0.1k & -0.1d \end{bmatrix} \\ \bar{p} &= \infty \end{aligned}$$

that is, we allow a maximum 10% reduction of the nominal values of the spring and damper elements. We do not impose upper bounds on the values of these elements. The (X, K, p) - Loop converges in 20 iterations with the numerical tolerance 0.1. The algorithm gives the optimal variation of p in the specified range: $\Delta p = [0.3810 \ -0.2000 \ 2.2559 \ 0.1709 \ 1.6491 \ 1.8916 \ 0.4726$

2.3944 0.5719 1.7181 0.8471 1.0662]. The controller gain is given by $G = [G_1 \ G_2]$, where

$$\begin{aligned} G_1 &= \begin{bmatrix} -1.3575 & -0.2822 & -1.1523 \\ -0.6162 & -0.2424 & -0.9776 \\ -0.9380 & -0.5166 & -3.9977 \end{bmatrix} \\ G_2 &= \begin{bmatrix} -1.5026 & -0.4545 & -2.5574 \\ -0.9091 & -0.5717 & -2.5769 \\ -1.0229 & -0.5154 & -4.8066 \end{bmatrix} \end{aligned}$$

The control energy in each iteration is plotted in Fig. 1. The impulse responses of the initial open-loop structure, redesigned open-loop structure, closed-loop system associated with the redesigned open-loop structure and the controller is given in Fig. 2.

Notice that even though we did not impose the upper-bound for p , the redesigned parameter values do not go to infinity. In fact, k_2 goes to the lower bound, and the rest parameters go to the optimal values shown above. The control effort is monotonically decreasing during the redesign process. Finally we have 75% control energy reduction.

9 Conclusions

In this paper, an algorithm for integrating structure and control design is presented. The overall optimization problem is divided into two subproblems with iteration between the two. The first subproblem gives the optimal values for the structure passive parameters (stiffness, damping ratios and actuator location parameters) within the prespecified range. The second subproblem gives the optimal values for the actuator location parameter and a controller with minimal control effort subject to the RMS constraints on the output. The algorithm OMPAC (Optimal Mix of Passive and Active Control) reduces control effort and convergence is guaranteed. For the future work, the output dynamic output feedback case will be considered, as well as mass minimization.

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Fig. 1 Control Energy during Redesign Process

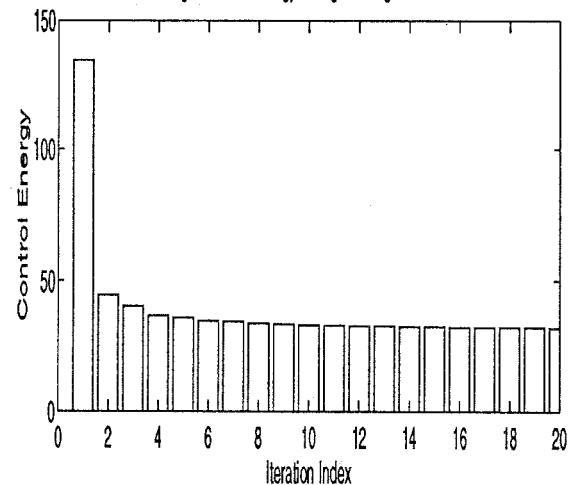


Fig. 2 Comparison of Impulse Responses

